We have already discussed about the multiplication of decimal numbers by 10, 100 and 1000. In the same way we can find patterns for multiplying decimal numbers by 0.1, 0.01 and 0.001. Observe the following.

$$12.3 \times 0.1 = \frac{123}{10} \times \frac{1}{10} = \frac{123}{100} = 1.23$$

$$12.3 \times 0.01 = \frac{123}{10} \times \frac{1}{100} = \frac{123}{1000} = 0.123$$

$$12.3 \times 0.001 = \frac{123}{10} \times \frac{1}{1000} = \frac{123}{10000} = 0.0123$$

From the above multiplication we can conclude that, when multiplying by

- 0.1, the decimal point moves one place left.
- 0.01, the decimal point moves two places left.
- 0.001, the decimal point moves three places left.

Zeros may be needed after the decimal point as they are needed before the decimal point of numbers when multiplied 0.1, 0.01 and 0.0001.

Try this

Complete the following table:

11.6×0.1	$\frac{116}{10} \times \frac{1}{10}$	–	–
11.6×0.01	–	–	–
11.6×0.001	–	–	–

Exercise 1.3

1. Find the product of the following
   (i) 0.5×3 (ii) 3.75×6 (iii) 50.2×4
   (iv) 0.03×9 (v) 453.03×7 (vi) 4×0.7
2. Find the area of the parallelogram whose base is 6.8 *cm* and height is 3.5 *cm*.
3. Find the area of the rectangle whose length is 23.7 *cm* and breadth is 15.2 *cm*.
4. Multiply the following
   (i) 2.57×10 (ii) 0.51×10 (iii) 125.367×100 (iv) 34.51×100
   (v) 62.735×100 (vi) 0.7×10 (vii) 0.03×100 (viii) 0.4×1000
5. A wheel of a baby cycle covers 49.7 *cm* in one rotation. Find the distance covered in 10 rotations.
6. A picture chart costs ₹1.50. Radha wants to buy 20 charts to make an album. How much does she have to pay?

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